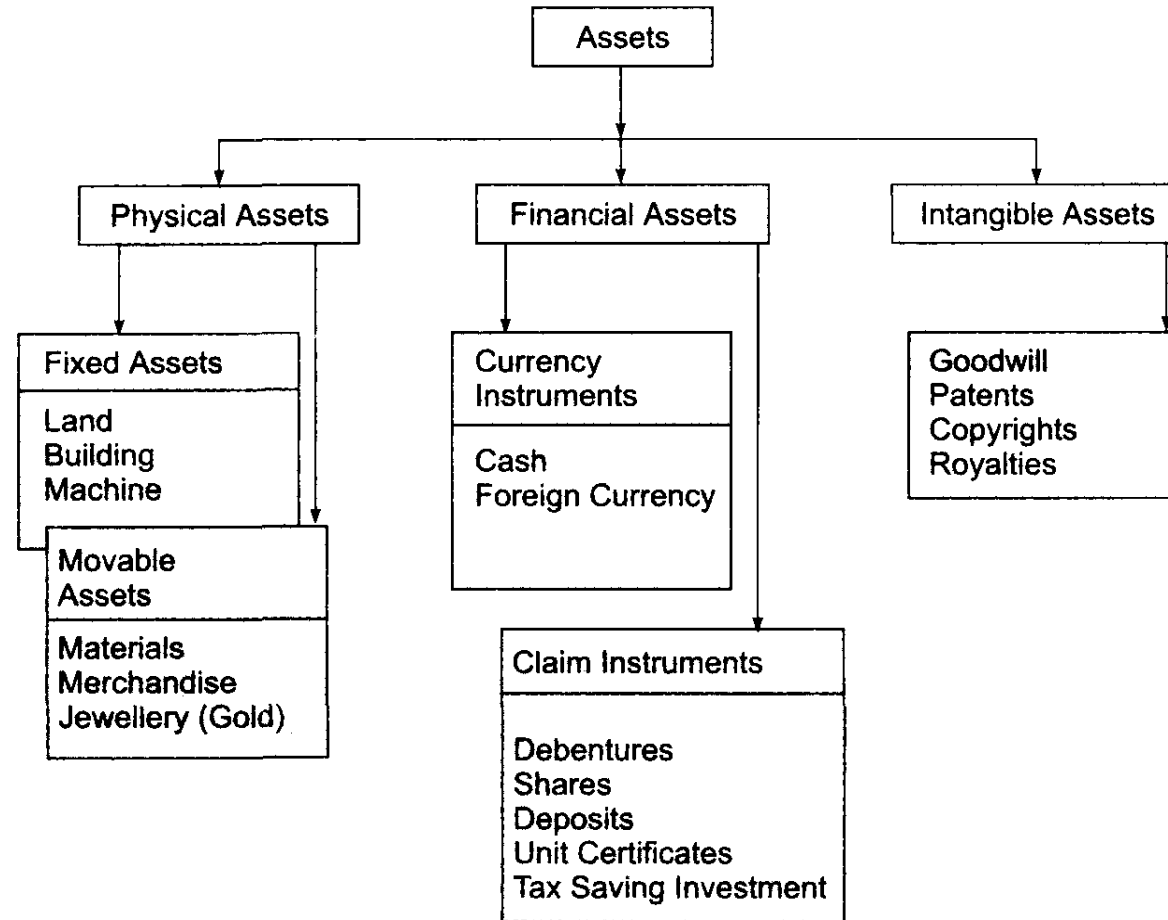


Investment Management

Asset Classification



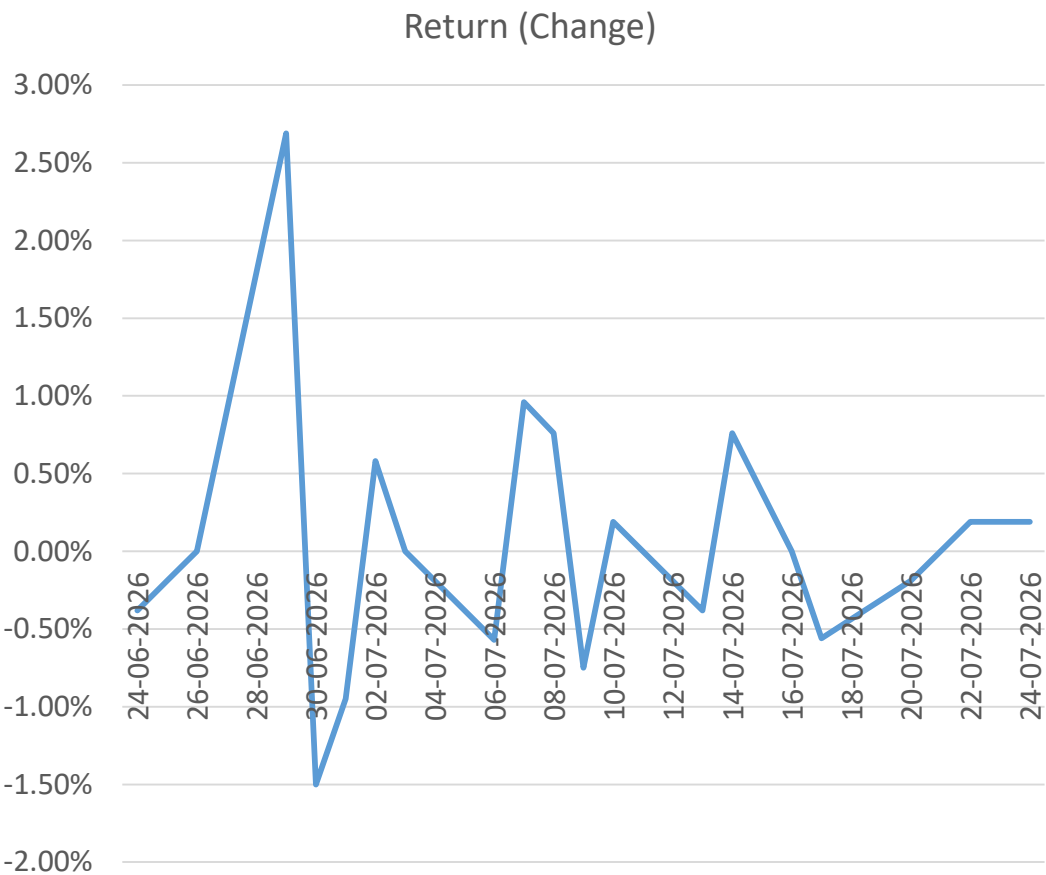
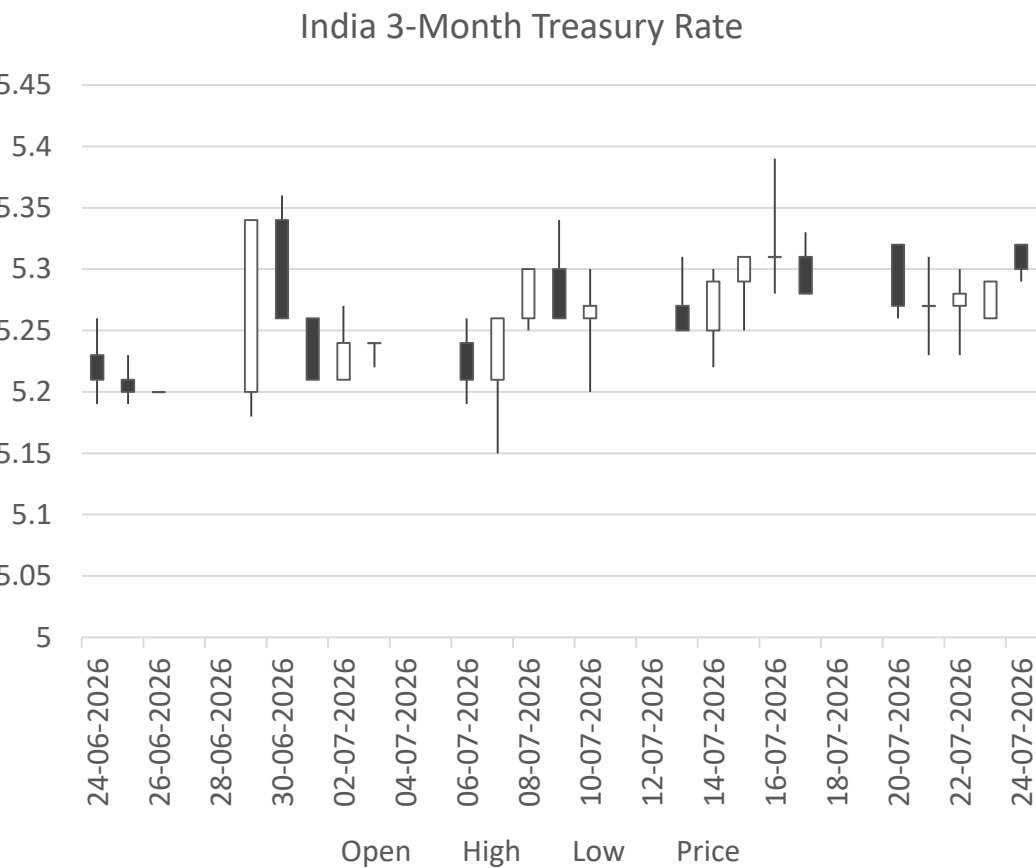
Characteristics of Investments

- Return
 - Bond coupon rate
 - Market return
 - Dividend return
- Risk
 - Standard deviation of return
 - Value at risk
- Safety
 - Credit rating
 - Investment rating
- Liquidity
 - Trading volume

Investment Risk – free Asset

- Markowitz approach: Investing for a single holding period, the return on the risk-free asset is certain.
- Terminal value expectations of the asset are not subject to uncertainty.
- Standard deviation of risk-free asset returns by definition is zero.
- Covariance between rate of return of risk-free asset and risky asset is zero.
 $\sigma_{ij} = \rho_{ij} \times \sigma_i \times \sigma_j$; if $\sigma_i = 0$; then $\sigma_{ij} = 0$
- Risk-free asset by definition has certain return ie., fixed income security with no possibility of default.
- All corporates are subject to default risk, therefore it is not possible for investor to have a risk-free asset issued by a firm.
- Government issue on the other hand is not subject to default, hence treasury instruments are considered as risk-free investments.

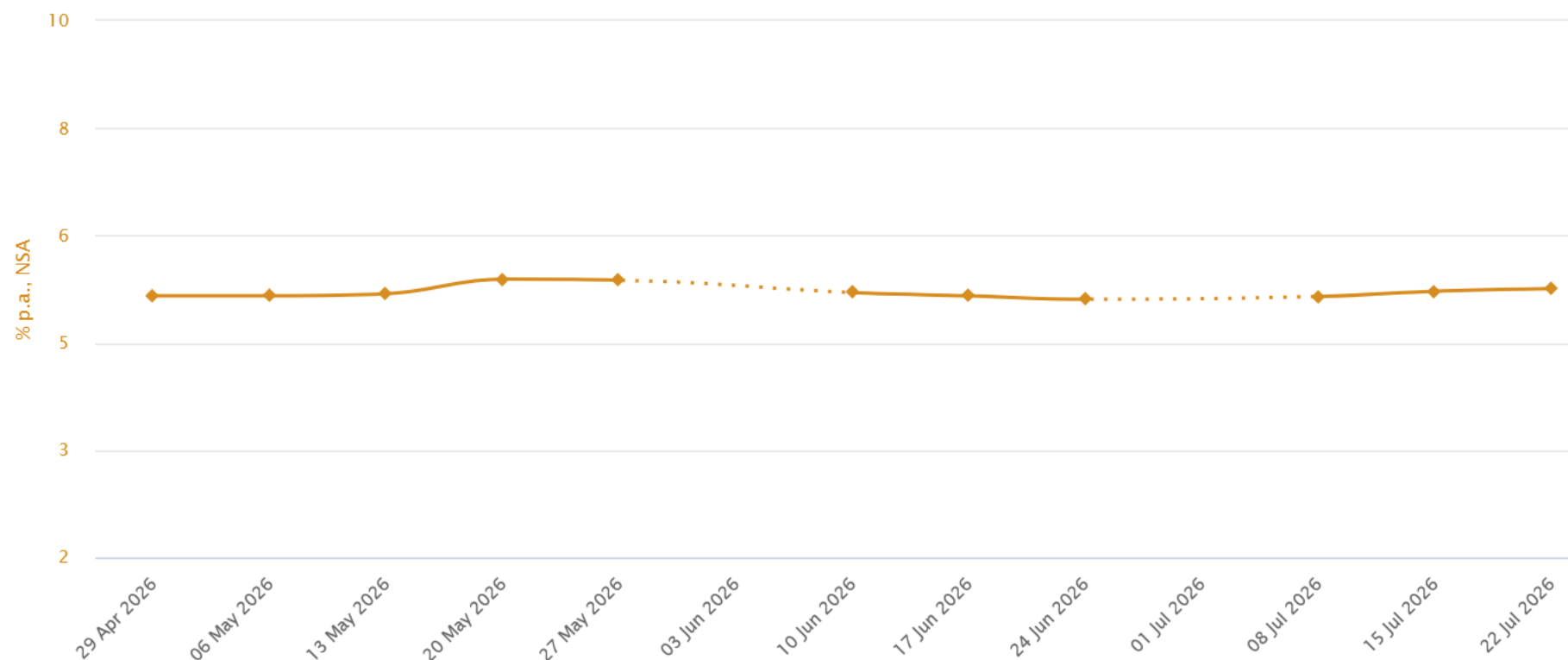
Risk-free Rate (India)



Standard deviation of daily return: 0.008137

Treasury Investment Risk

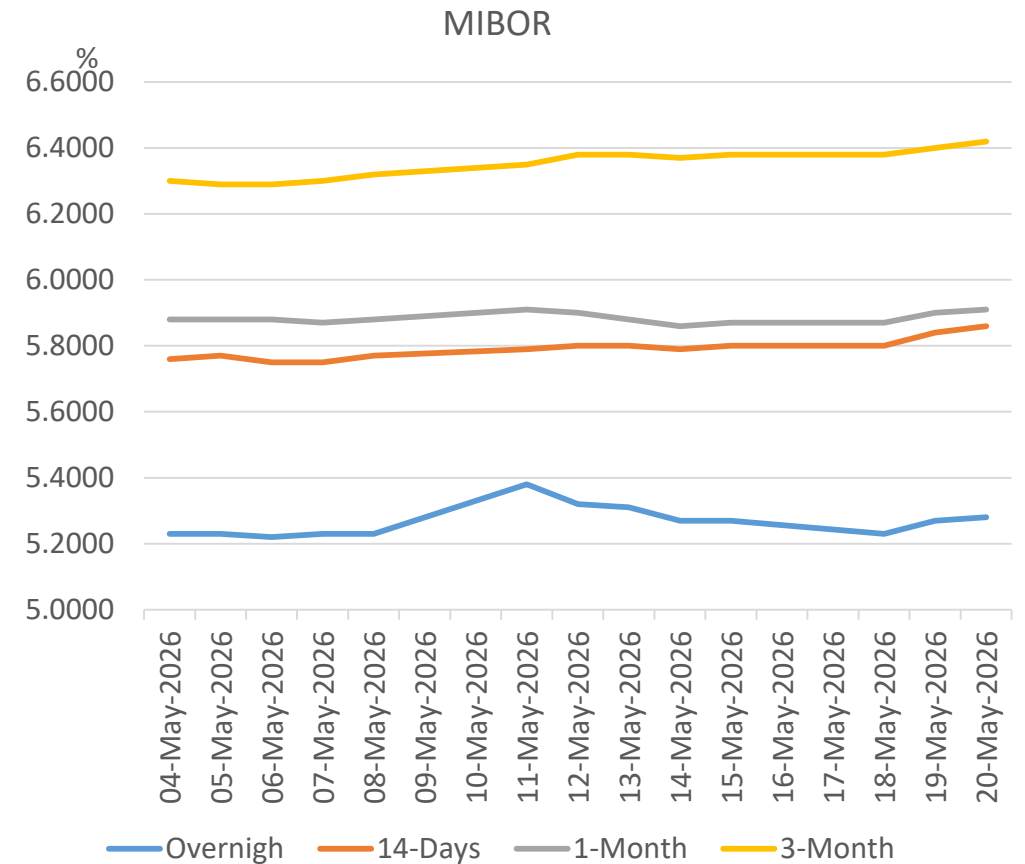
- Investment holding period: One Month



Source: Moody's analytics: <https://www.economy.com/india/treasury-bills-over-31-days>

Treasury Investment Risk

- Investment Holding Period: 12 Days:
- Risk-free rate: 1-Month MIBOR
 - Return is uncertain implying **price risk** of investment
- Risk-free rate: 14-Days MIBOR
 - Reinvestment requirement for 1 day implies **reinvestment risk**
- Risk-free rate: Overnight MIBOR
 - **Risk-free rate**

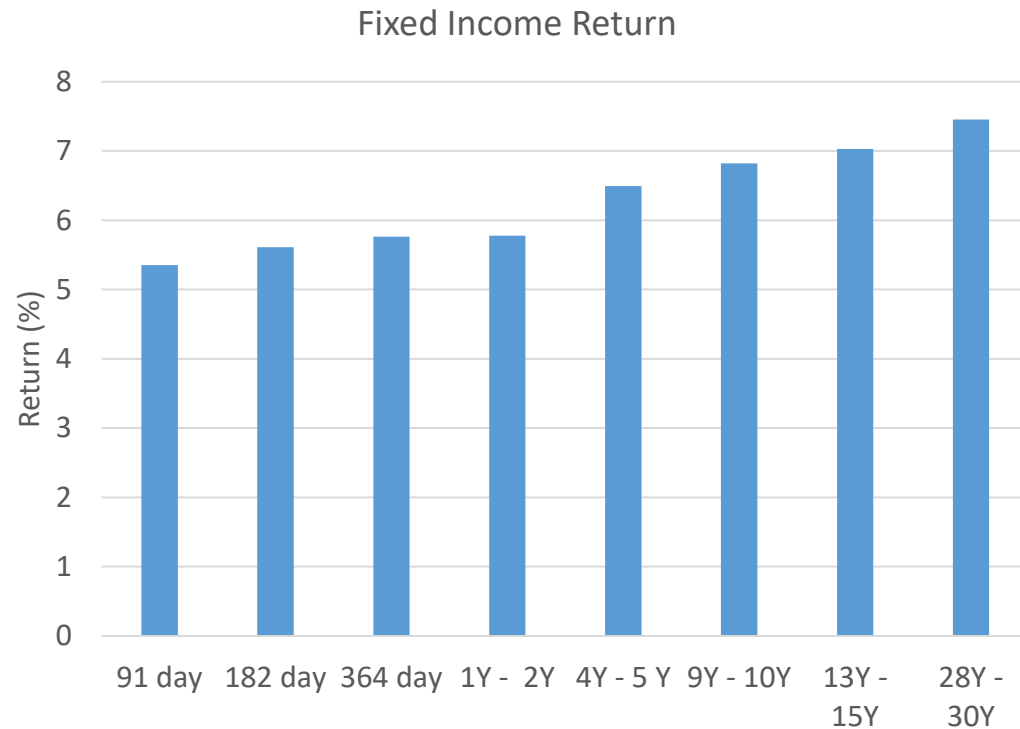


Objectives of Investment

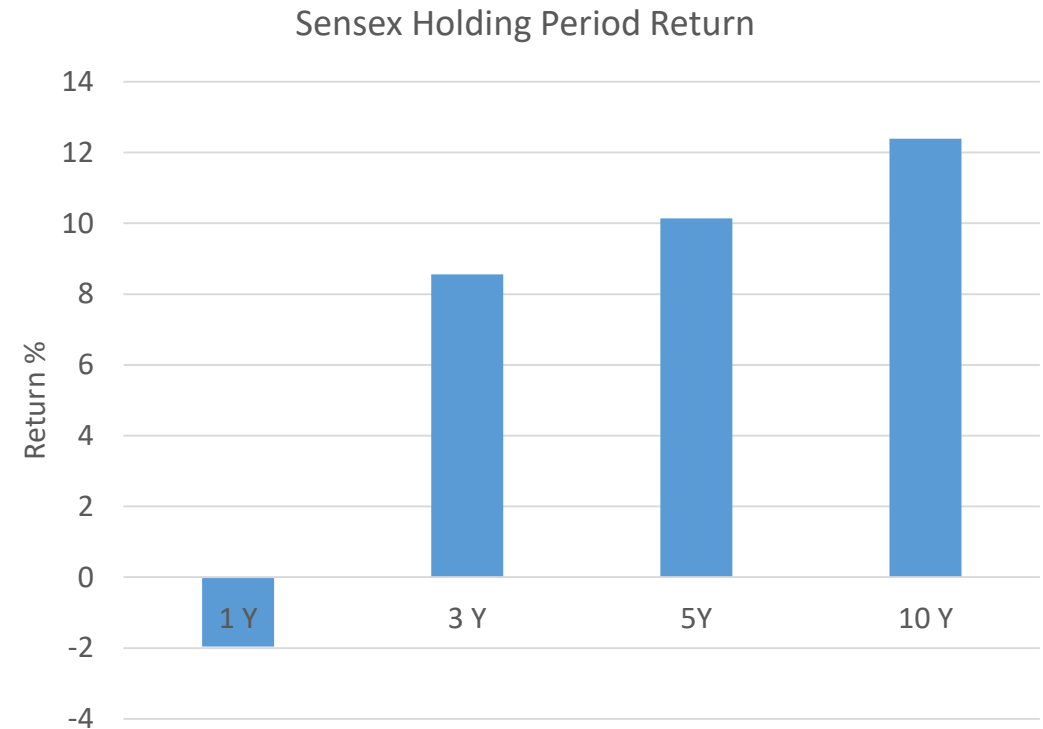
- Risk – Return balance
 - Preference for mutual funds if it offers higher than market return with moderate risk
 - Investment in IPOs when risk is perceived to be high would be made only when return expectation is exceptionally high
- Safety and Liquidity
 - Income yielding assets than growth yielding assets would be preferred by retired employees
 - Expected return from a financial asset is higher than the prevalent inflation rate
- Utilization of tax incentives
 - Reduce present tax payment and achieve a marginal rate of return on investments
 - Preference for provident fund / insurance schemes to reduce tax

Market Returns

Interest Income



Capital Gains



Inflation rate: 4.38%; Interest rate: 5.25% Source: <https://www.ccilindia.com/tenorwise-indicative-yields#>

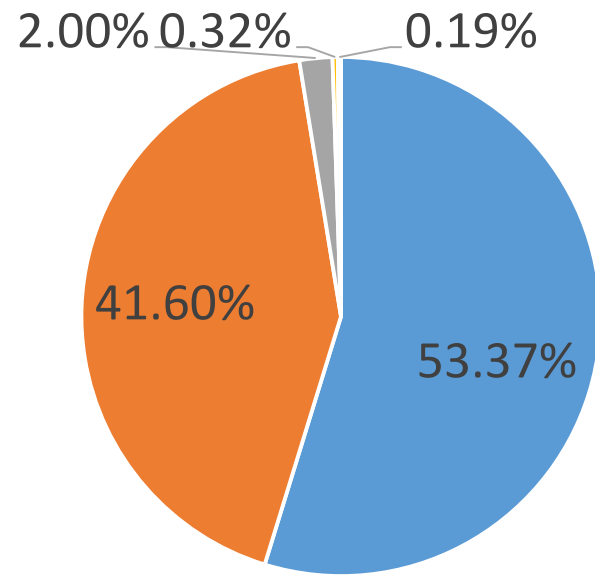
Investment Model

- Diversification implies when securities are combined as a portfolio, portfolio will exhibit lower risk than the simple average risk of the combined securities.
- Security returns are combined as a portfolio. Portfolio return is the weighted average return of the combined securities.
- Increased returns are hence matched by reduced risk when securities are diversified in a portfolio.

Investment Portfolios

- Debt Portfolios
 - Government securities
 - Corporate securities
 - Alternative Investments: Private equity, real estate, venture capital, commodities
 - Pass through Certificates: Securitized debt instruments (Proportional ownership interest in a pool of underlying assets, such as mortgages, auto loans, or microfinance receivables). Investors receive regular proportionate payments of principal and interest as borrowers pay off their loans.
- Equity Portfolios
- Hybrid Portfolios

Debt Portfolio Composition (Example)

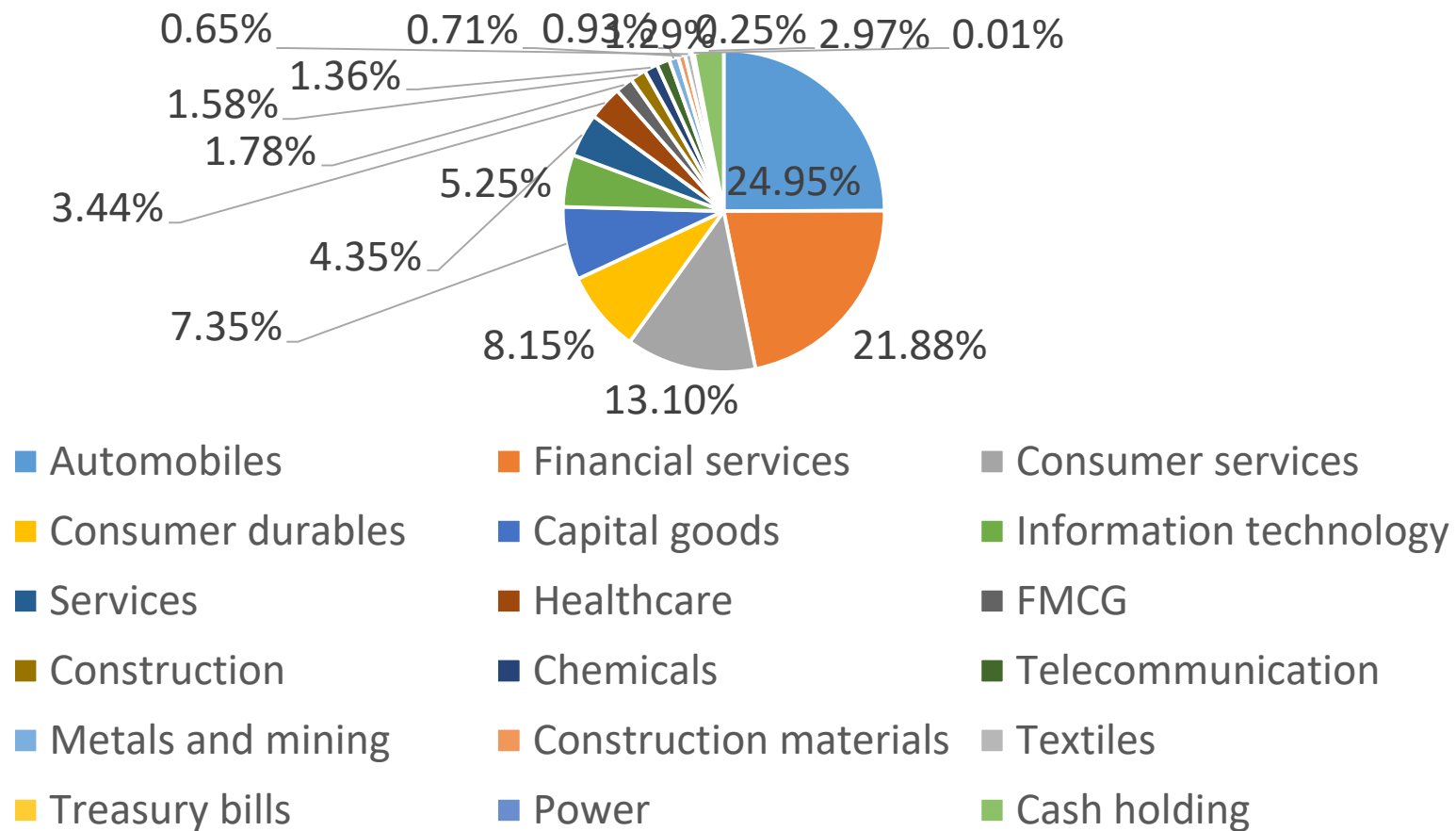


- Government Securities: Long term
- Corporate Securities
- Government Securities: Short term
- Alternate Investments
- Pass through Certificates

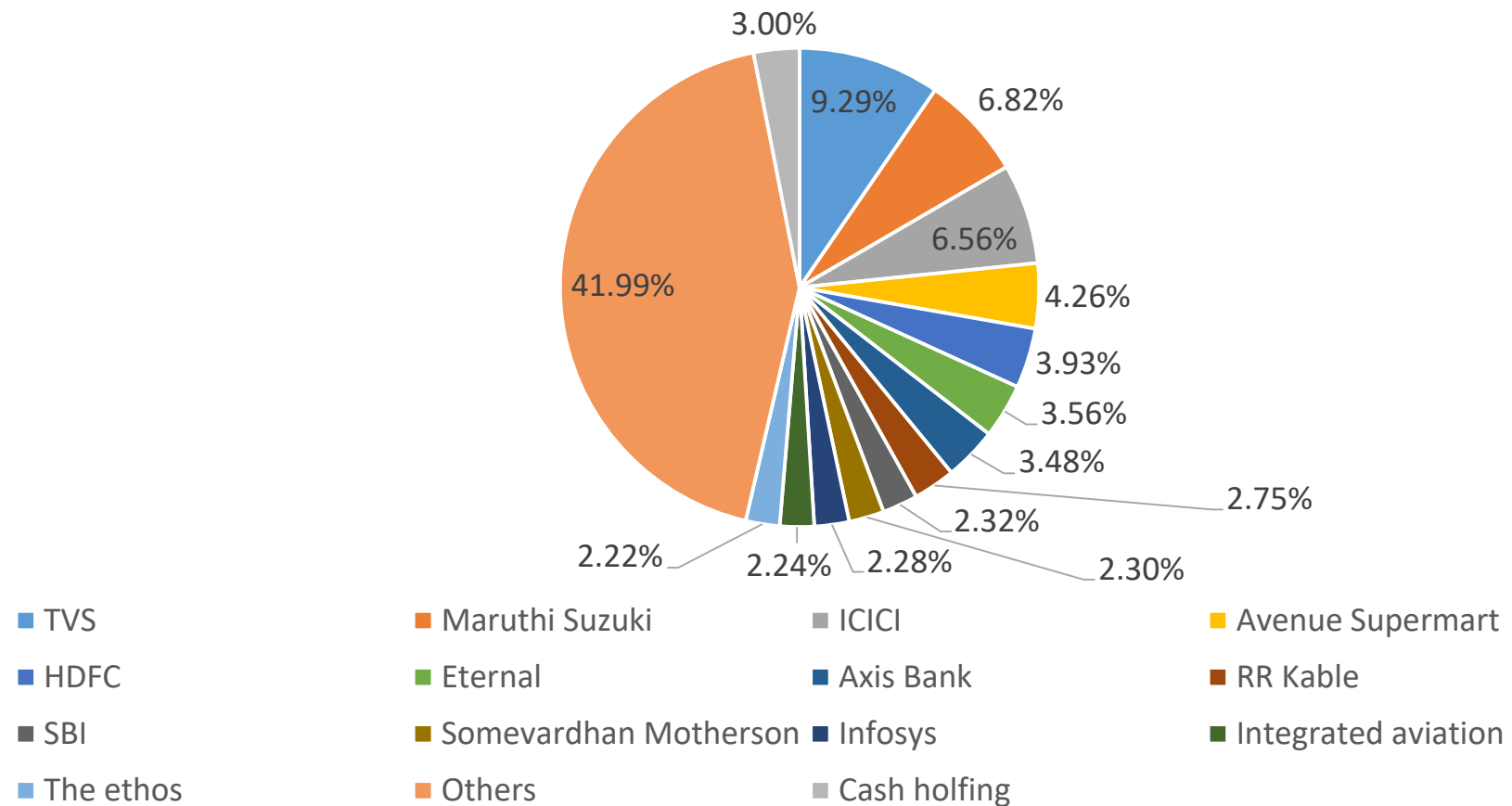
Investment Portfolios

- Debt Portfolios
- Equity Portfolios
 - Sector Portfolios: Manufacturing, energy, Information technology...
 - Business Cycle Portfolios: Shift investments based on the business cycle (high growth / stable growth)
 - Momentum Portfolios: Equities showing persistent price trends
 - Dividend yield Portfolios
 - Tax saving Portfolios
 - Minimum variance Portfolios
 - Large / Mid Cap / Small Cap Portfolios
 - Quant Portfolios: Applying mathematical models / algorithms
 - MNC Portfolios
- Hybrid Portfolios

Equity Portfolio Composition (Example)



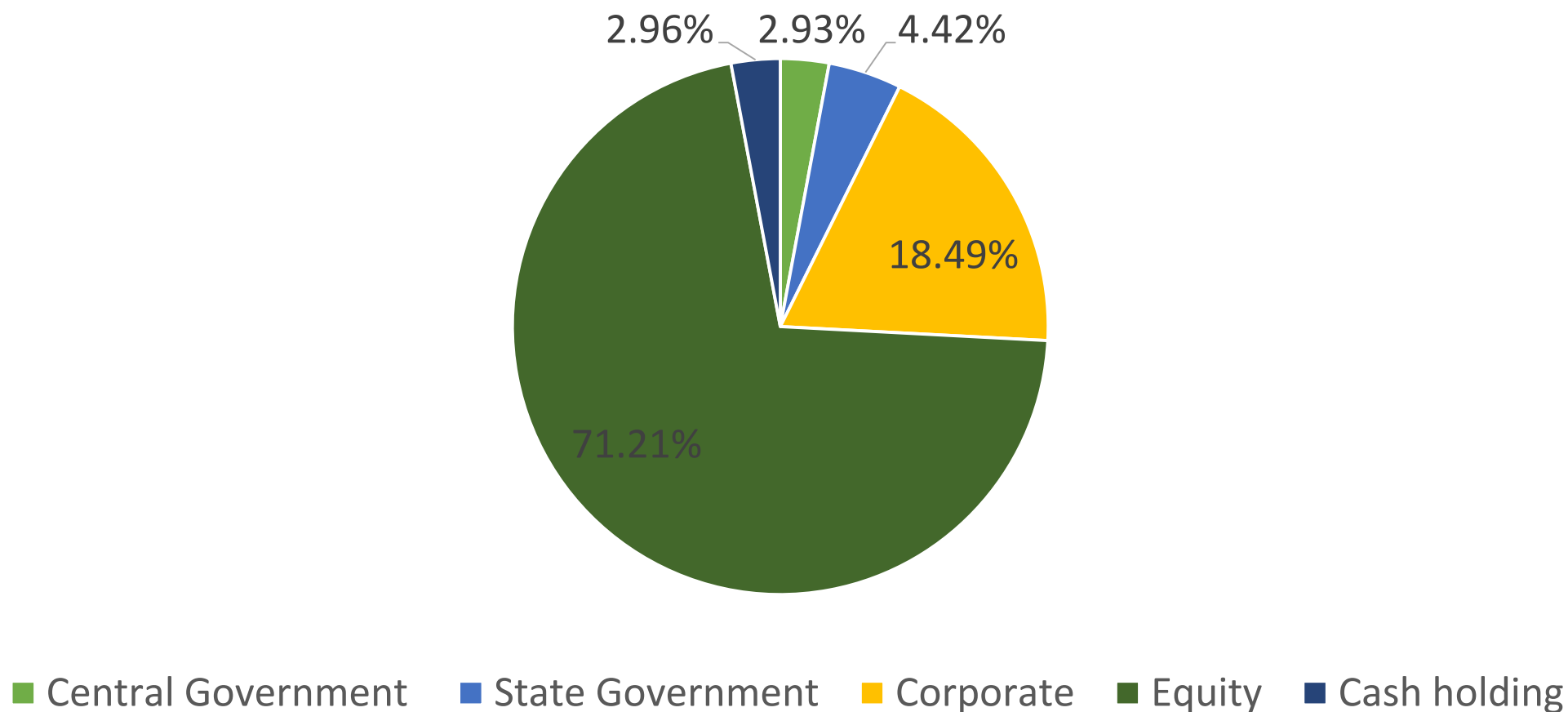
Equity Portfolio Composition: Top Holding (Example)



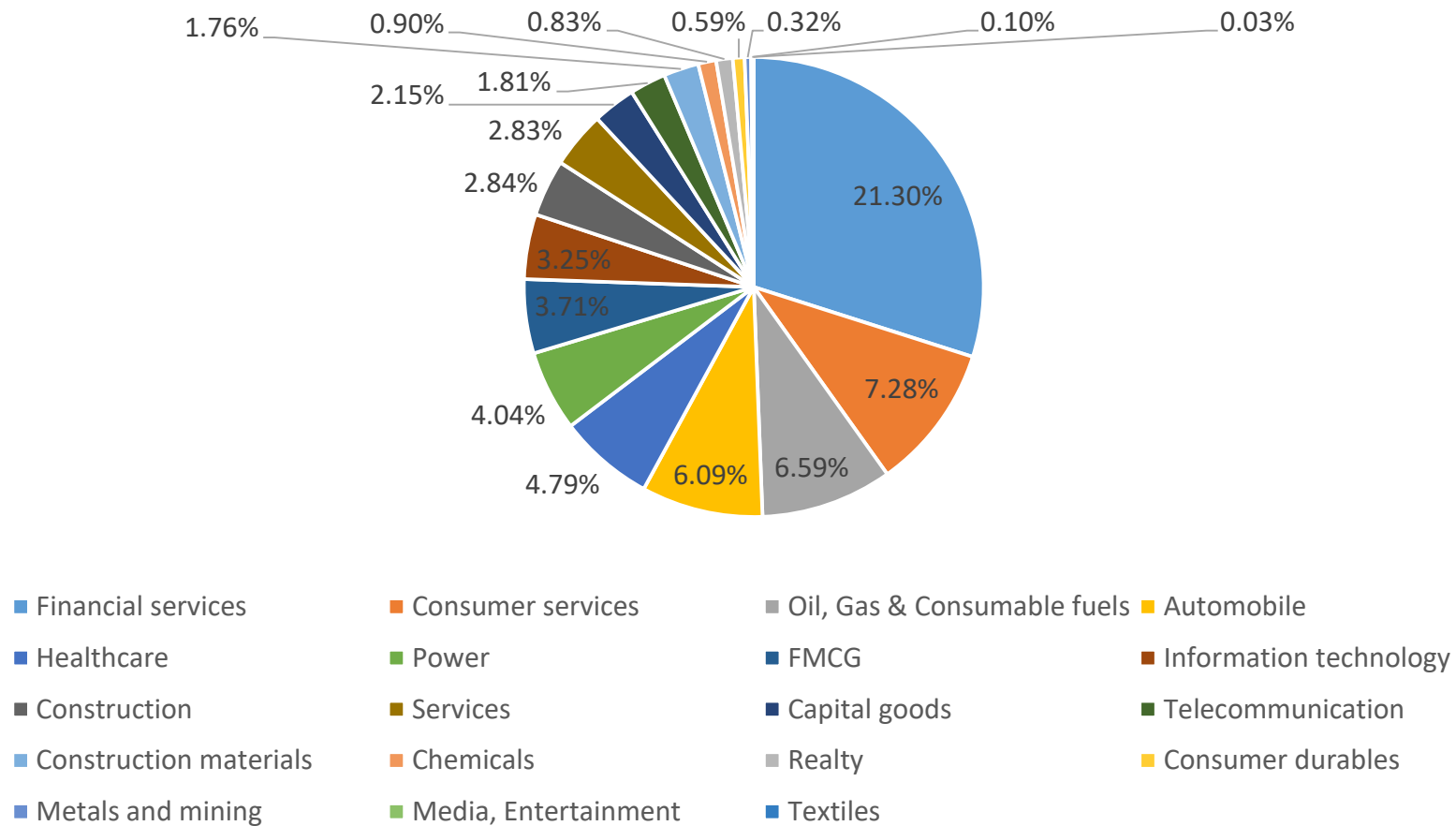
Investment Portfolios

- Debt Portfolios
- Equity Portfolios
- Hybrid Portfolios
 - Equity and Debt
 - Equity and Savings
 - Balanced portfolios
 - Multi Asset
 - Arbitrage portfolios

Debt (30%) Equity (70%) Portfolio (Example)



Debt (30%) Equity (70%) Portfolio (Example)



Evolution of Portfolio Management

- 'The Anatomy of a Railroad Report' by Thomas F. Woodlock (1900)
- 'The Art of Wall Street Investing', by John Moody (1906)
- 'The Principles of Bond Investment' by Lawrence Chamberlain (1911)
- Series of editorials in the Wall Street Journal by Charles H. Dow, during 1900-1902 in U.S.A.
- 'The Wave Principle' by Ralph N. Elliot (1938)
- 'Portfolio Selection' by Harry Markowitz in Journal of Finance (1952)

Phases of Portfolio Management

➤ **Speculative phase**

- Risk bearing capacity
 - Investors : Low
- Return expectations
 - Investors : Return expectations
- Duration of trade
 - Investors : Long

Speculators : High

Speculators: Capital gain expectations

Speculators : Short

➤ **Phase of Professionalism**

- Experiments and Evaluation

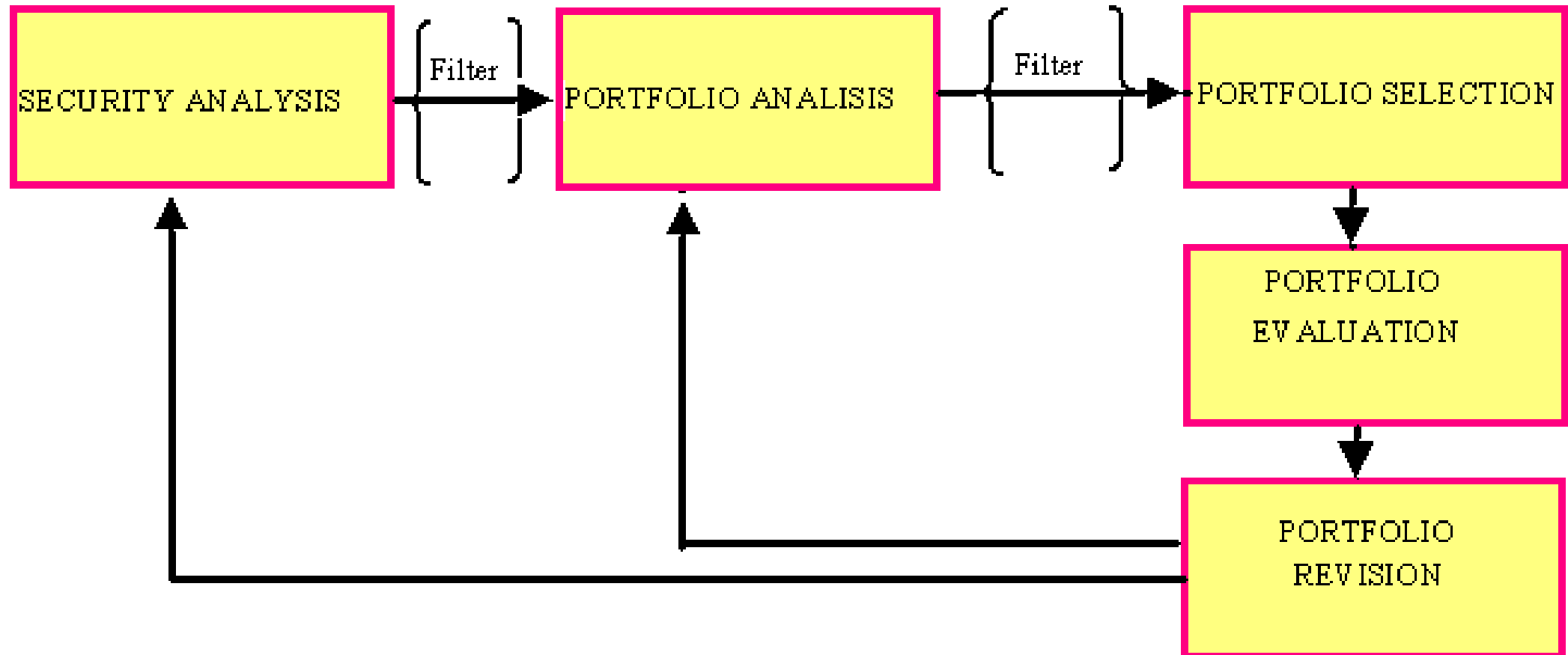
➤ **Scientific phase**

- Security analysis
- Portfolio analysis
- Portfolio selection
- Portfolio evaluation
- Portfolio revision

Investment Process

- Investment Objectives
- Analyze Securities
- Construct Investment Portfolio
- Evaluate the Portfolio performance
- Portfolio Revision

Portfolio Management Process



Class Exercise

Group 1	Group 2	Group3
Elcid Investment	Swan Defence	Taparia Tools
Diamond Power	Cupid	KSE
Swan Defence	Modern Insulator	PTL Enterprises
Midwest Energy	Sterlite Technologies	Coal India
SG Finserv	Sigma Advanced System	D-Link India
Reliance Industries	TCS	Maruti Suzuki
Bharti Airtel	Bajaj Finance	Infosys
HDFC Bank	Larsen & Toubro	Titan Company
ICICI Bank	Hind. Unilever	Adani Ports
SBI	Sun Pharma.Inds.	M & M

Class Exercise

[illegible]